JONNAKUTI SHARATH CHANDRA

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PROFESSIONAL SUMMARY

B.Tech graduate in Electrical, Electronics and Communications Engineering with hands-on experience in Embedded Systems, Robotics, IoT-based projects, R&D projects, and technical mentoring. Skilled in Embedded C, C++, Arduino, microcontrollers, sensors, actuators, communication modules, and real-time hardware interfacing. Experienced in developing, testing, and troubleshooting embedded and robotics projects, while mentoring students in technology-based projects and practical learning. Also completed training in RTL Design and Verification using Verilog and SystemVerilog. Interested in opportunities in Embedded Systems, Firmware Development, R&D, Robotics, IoT, and Hardware Design.

PROFESSIONAL EXPERIENCE

Robotics Trainer & Technical Mentor — Leap Robots, Hyderabad 09/2025 – Present
Hyderabad

- Conduct hands-on training in **Robotics, Embedded Systems, IoT, AI, and coding** for students through practical, project-based learning.
- Design, develop, test, and troubleshoot **R&D and embedded-system projects** involving microcontrollers, sensors, actuators, motors, servos, and communication modules.
- Develop and implement real-time projects using **Arduino, ESP32, ESP8266, and other embedded platforms**.
- Mentor students in project development, helping them understand **circuit design, programming, hardware interfacing, testing, and debugging**.
- Guide students in developing innovative projects and prepare them for **project demonstrations, competitions, and exhibitions**.
- Conduct technical sessions and provide support in **problem-solving, debugging, and improving project functionality**.
- Work on projects integrating **AI, robotics, IoT, sensors, and automation** to create practical solutions.
- Collaborate with students and team members during **R&D, prototyping, testing, and project implementation**.

Training Experience — KT Semicon, Hyderabad 02/2025 – 09/2025
Hyderabad

- Completed training in **RTL Design.**
- Worked with **Verilog, SystemVerilog, digital design, HDL simulation and debugging, logic synthesis, and timing analysis**.

EDUCATION

Bachelors of Technology-[ECE] 11/2021 – 05/2025
Seshadri Rao Gudlavalleru Engineering College Gudlavalleru
CGPA-8.3

MPC 07/2019 – 03/2021
Sri Chaitanya Junior College Guntur
Percentage-75%

Zilla Parishad High School 07/2018 – 05/2019
CGPA-8.3 Mandepudi

SKILLS

Programming
- Embedded C
- C++
- Microcontroller Programming

Soft Skills
- Problem Solving
- Teamwork & Collaboration
- Communication
- Creativity
- Leadership
- Time Management
- Adaptability
- Technical Training & Mentoring

Hardware Skills
- Circuit Design
- Sensor Integration
- Embedded Hardware Debugging
- Electronic Components
- Hardware Testing & Troubleshooting

Embedded Systems
- Arduino
- ESP32
- ESP8266
- Microcontroller Programming
- GPIO & Hardware Interfacing
- Sensors & Actuators
- Serial Communication
- Motor & Servo Control
- IoT

LANGUAGES

Telugu
Native

Hindi
Fluent

English
Proficient

CERTIFICATES & WORKSHOPS

NPTEL – Introduction to Internet of Things

Programming with Python

PCB Design Workshop Gudlavalleru
Learned schematic design, PCB layout.

IoT Workshop
Tinkercad
Learned IoT concepts and implemented basic simulations using Tinkercad software

PROJECTS

Moisture Sensing Automatic Plant Watering System
- Built an Arduino-based system with soil moisture sensors for automatic irrigation.
- Automated pump control to reduce manual effort and prevent overwatering.
- Enabled real-time sensing and actuation for improved efficiency.

IoT-Based Smart Home Automation
- Developed a smart home automation system using **ESP8266/ESP32 and sensors**.
- Controlled electrical appliances using relay modules and monitored sensor data.
- Implemented IoT connectivity for remote monitoring and control.

Pan & Tilt Camera Car

- Developed a **Pan & Tilt Camera Car** using a microcontroller, DC motors, motor driver, and servo motors.
- Implemented **wireless control of the car's movement** with forward, backward, left, and right motion.
- Integrated a **pan-and-tilt camera mechanism** using servo motors to provide adjustable camera viewing angles.
- Focused on **motor control, servo control, hardware interfacing, and real-time embedded system operation**.

Buddy Band – Smart Safety & Assistance Wearable

- Developed a **wearable smart band** designed to provide quick assistance through predefined emergency and location-based alerts.
- Integrated **OLED display, push buttons, vibration motor, and DFPlayer Mini** for user interaction and audio feedback.
- Implemented predefined location selection and alert functionality without relying on GPS.

- Focused on **embedded programming, hardware interfacing, user interaction, and real-time response**.

Sensor Recognition Candy Machine with AI

- Developed an **AI-enabled candy dispensing machine** that recognizes sensor names provided by the user.
- When the entered sensor name is **correct**, the system indicates it with a **green LED**, activates the **servo motor**, and dispenses a candy.
- If the entered sensor name is **incorrect**, the system indicates the error using a **red LED** and does not dispense the candy.
- Integrated a **microcontroller, AI-based recognition, LEDs, and servo motor** to create an interactive real-time learning system.

Rock Paper Scissors Game

- Developed an interactive **Rock Paper Scissors game** using a microcontroller.
- Integrated an **LCD display** to show the player's choice, system choice, and game result.
- Used a **push button** to allow the player to make a selection and control the game.
- Programmed the microcontroller to determine the winner based on the selected choices and display the result in real time.

EXTRA-CURRICULAR ACTIVITIES

Electronics and Communication Students Association (ECSAt) 07/2024 – 05/2025
President Gudlavalleru

- Led a 20+ member team, managing technical and cultural events.
- Launched mentorship programs with alumni and industry experts.
- Encouraged participation in technical competitions, resulting in multiple awards.

Electronics and Communication Students Association (ECSAt) 07/2023 – 06/2024
Vice President Gudlavalleru

- Directed a 15-member team to organize college-level tech fests, boosting participation by 40%.
- Secured sponsorships and increased funding by 20%.
- Conducted training sessions on VHDL, IoT, Embedded Systems, and Web Development.